AI-Supported Design Process Summary

Sherbet Machine Design Project

Keywords

- Compact Living
- Mobility
- User Rituals

Selected Impactful Prompt

Example prompt used in the process:

Prompt

"Act as a **clarifier.** I am designing an electronic hot beverage preparation device for mobility-focused or compact living environments. Ask me 10 precise questions to help define the user's real context."

This prompt helped structure the early research phase and guided the definition of the design problem.

Initial Design Direction – Student-Oriented Beverage Device

At the beginning of the project, the design problem focused on creating a hot beverage preparation device for students living in dormitories or shared apartments.
The main goal was to understand how students prepare hot beverages in compact living environments with limited space and divided attention. The design aimed to address issues such as:

- limited counter space
- multitasking behavior
- low attention during use
- safety risks related to boiling water.

AI was used as a **clarifier** to generate potential user profiles and usage scenarios for students living in small dormitory rooms and shared kitchens.

Why I Abandoned the Student Context

Although the student-focused research helped me understand compact living environments, I later realized that the concept was becoming too generic.

Designing another kettle-like hot beverage device for students did not offer a strong cultural or conceptual differentiation. Many products already exist that address similar needs such as:

- fast boiling
- compact size
- simple interfaces.

Because of this, the project risked becoming a typical small kitchen appliance rather than a distinctive design proposal.
For this reason, I decided to move away from the student-centered direction and search for a more meaningful cultural context.

Keywords

- Sherbet culture
- Ottoman hospitality
- Hospitality experience
- Cultural storytelling

Selected Impactful Prompt

Example prompt used in the process:

Prompt

“I am designing a sherbet machine that will be used as a ritual serving object for guests. The machine should reinterpret Ottoman culture in a modern design language.”

This prompt helped expand the design exploration and refine the concept.

Design Pivot – Discovering Sherbet Culture

After exploring historical beverage traditions, I became interested in sherbet, a drink deeply connected with Ottoman hospitality and coffeehouse culture.
Unlike simple hot drinks prepared by students, sherbet involves a multi-stage preparation process including boiling, infusion, filtering, and sweetening.

This complexity revealed a clear design opportunity:
creating a machine that could **automate the preparation process** while preserving the cultural ritual.

Research Phase – Sherbet Production

Further research showed that existing sherbet machines only store and display the beverage, but they do not produce it.

This insight led to the key opportunity:
Integrating sherbet production and storage in a single machine.
Another important realization was that sherbet preparation is not only functional but also **experiential and symbolic**.

Concept Development

Based on these insights, the design evolved into a sherbet preparation machine that:

- prepares sherbet automatically
- filters ingredients
- preserves the beverage
- allows guests to observe the preparation process.

This approach transforms the product from a simple appliance into a **ritual-oriented hospitality object**.

Selected Impactful Prompts

Example prompt used in the process:

Prompts

Prompt 1 – Design Critique

"A friend of mine designed a sherbet preparation machine, but I am unsure about the design. From a designer's perspective, what are the weaknesses of this machine? Which aspects should be reconsidered or redesigned? Please evaluate the concept critically and explain what improvements could make the design stronger."

Prompt 2 – Reframing the Design

"After discussing the design again with my friend, we revised the machine accordingly. Could you evaluate the updated concept from a reframing perspective? Are there still missing elements? Is the system realistically producible? I am still uncertain about the feasibility of the machine, so please analyze the design and suggest possible improvements."

Ideation and Prototype

After defining the concept of a sherbet preparation machine, I began the ideation phase by producing quick sketches to explore the possible form and internal structure of the device. The goal was to understand how the different stages of sherbet preparation—such as boiling, filtering, and serving—could be integrated into a single system.

During this process, I realized that my first concept was too complex and the internal mechanism was not clearly understandable. The system contained too many layers and did not resemble a realistically producible machine. Therefore, I decided to redesign the structure and simplify the system logic to create a clearer and more feasible design direction. Then, based on the feedback I received from the artificial intelligence, I improved the system, created a prototype, and finalized the product.

AI-Supported Exploration

Artificial intelligence tools such as ChatGPT, Vizcom, and Gemini were used during the ideation and exploration phases to support the design process.

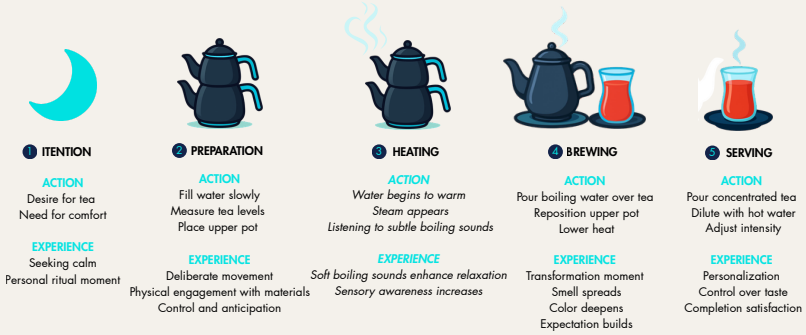
AI contributed in several ways:

- generating alternative form explorations
- visualizing possible product proportions
- helping reinterpret the design language
- supporting iterative concept development.

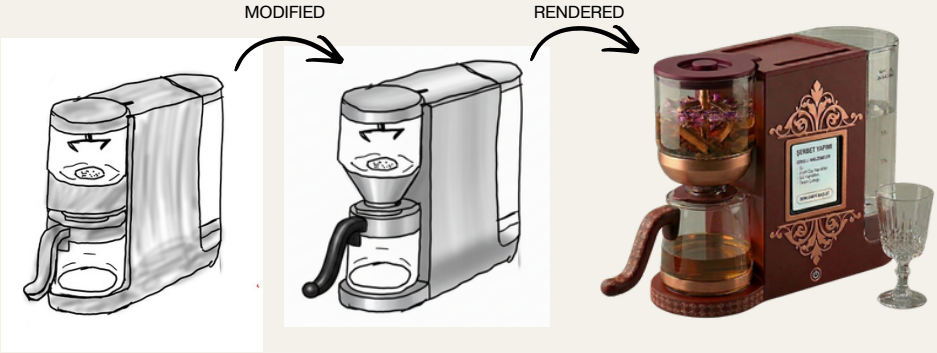
AI-Supported Design Timeline

Sherbet Machine Design Project

Initial Design Direction – Student-Oriented Beverage Device



The first design of the sherbet machine.



First Concept

inspiration

Second Concept

inspiration

First Concept Selection and Detailed Development

Final Design